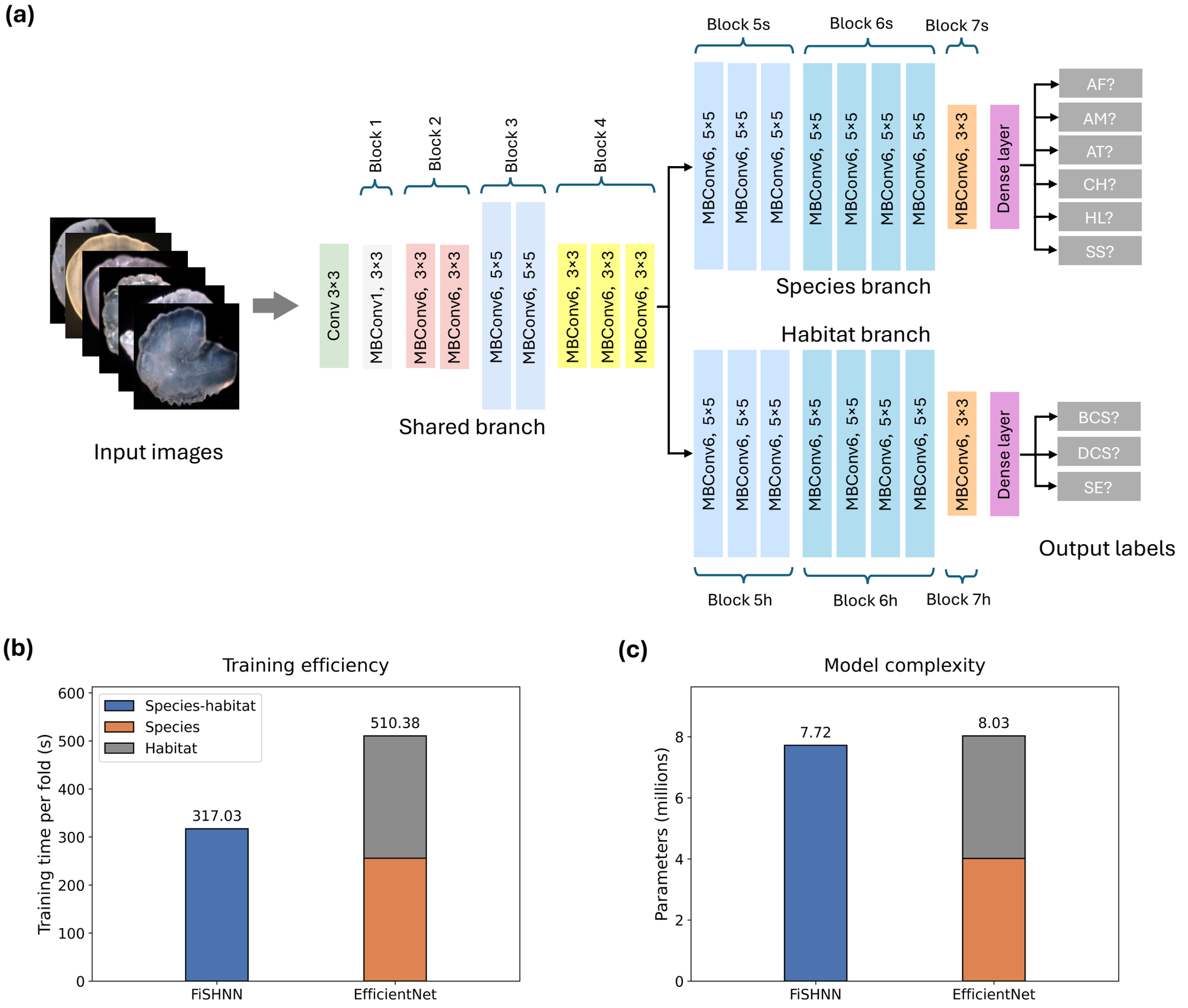
(a)
Input images
Conv 3×3
MBConv1, 3×3
MBConv6, 3×3
MBConv6, 5×5
MBConv6, 3×3
Block 1
Block 2
Block 3
Block 4
Shared branch
Block 5s
Block 6s
Block 7s
Species branch
Habitat branch
Dense layer
AF?
AM?
AT?
CH?
HL?
SS?
BCS?
DCS?
SE?
Output labels
Block 5h
Block 6h
Block 7h
(b)
Training efficiency
Training time per fold (s)
Species-habitat
Species
Habitat
317.03
510.38
FiSHNN
EfficientNet
(c)
Model complexity
Parameters (millions)
7.72
8.03
FiSHNN
EfficientNet